

Instructions

This mock mirrors the BI3 flow: aptitude + reasoning, then DE technical MCQs, then SQL and coding.
This is Mock 1 of 3; each uses a different question set.

Section 1 - Aptitude

Q1. What is 25% of 240?

(A) 50

(B) 60 ✓

(C) 70

(D) 80

Answer: (B) 60

Authenticity: Community Reported

Q2. Average of 12, 18 and 24:

(A) 16

(B) 18 ✓

(C) 20

(D) 22

Answer: (B) 18

Authenticity: Community Reported

Q3. SI on Rs. 1000 at 6% for 2 years:

(A) Rs. 100

(B) Rs. 120 ✓

(C) Rs. 60

(D) Rs. 200

Answer: (B) Rs. 120

Authenticity: Community Reported

Q4. Divide Rs. 500 in the ratio 2:3. Larger share:

(A) Rs. 200

(B) Rs. 300 ✓

(C) Rs. 250

(D) Rs. 350

Answer: (B) Rs. 300

Authenticity: Community Reported

Q5. Two numbers in ratio 5:7 sum to 72. Smaller:

(A) 30 ✓

(B) 35

(C) 28

(D) 32

Answer: (A) 30

Authenticity: Community Reported

Q6. A:B = 2:5, B = 25. Then A is:

(A) 10 ✓

(B) 12

(C) 15

(D) 8

Answer: (A) 10

Authenticity: Research-Backed Company Style

Section 2 - Reasoning

Q1. Next term: 3, 6, 9, 12, ?

(A) 14

(B) 15 ✓

(C) 16

(D) 18

Answer: (B) 15

Authenticity: Frequently Repeated

Q2. Finger : Hand :: Leaf : ?

(A) Tree

(B) Branch ✓

(C) Root

(D) Flower

Answer: (B) Branch

Authenticity: Community Reported

Q3. If RED = 27 (18+5+4), then BLUE (2+12+21+5) is:

(A) 38

(B) 40 ✓

(C) 42

(D) 36

Answer: (B) 40

Authenticity: Community Reported

Q4. Odd one out: 2, 3, 5, 9, 11.

(A) 3

(B) 5

(C) 9 ✓

(D) 11

Answer: (C) 9

Authenticity: Community Reported

Q5. Walk 3 km east then 4 km north. Distance from start:

(A) 5 km ✓

(B) 7 km

(C) 6 km

(D) 1 km

Answer: (A) 5 km

Authenticity: Community Reported

Section 3 - Data Engineering Technical

Q1. Which clause filters rows AFTER aggregation?

(A) WHERE

(B) HAVING ✓

(C) GROUP BY

(D) ORDER BY

Answer: (B) HAVING

Authenticity: Frequently Repeated

Q2. Which normal form removes partial dependency on part of a composite key?

(A) 1NF

(B) 2NF ✓

(C) 3NF

(D) BCNF

Answer: (B) 2NF

Authenticity: Community Reported

Q3. The central table holding measures is the:

(A) Dimension table

(B) Fact table ✓

(C) Lookup table

(D) Bridge table

Answer: (B) Fact table

Authenticity: Community Reported

Q4. OLAP systems are optimized for:

(A) High-volume writes

(B) Analytical queries and aggregation ✓

(C) Row locking

(D) Payments

Answer: (B) Analytical queries and aggregation

Authenticity: Community Reported

Q5. ETL stands for:

(A) Extract, Transform, Load ✓

(B) Export, Transfer, Link

(C) Extract, Transfer, Log

(D) Evaluate, Transform, Load

Answer: (A) Extract, Transform, Load

Authenticity: Frequently Repeated

Q6. Apache Spark processes data mainly via:

(A) Disk-only MapReduce

(B) In-memory distributed computing ✓

(C) Single-threaded loops

(D) Stored procedures

Answer: (B) In-memory distributed computing

Authenticity: Community Reported

Q7. Snowflake separates storage and compute so they:

(A) Scale together

(B) Scale independently ✓

(C) Are disabled

(D) Are limited

Answer: (B) Scale independently

Authenticity: Community Reported

Q8. A columnar file format common in analytics:

(A) CSV

(B) Parquet ✓

(C) TXT

(D) XML

Answer: (B) Parquet

Authenticity: Community Reported

Q9. Library most used for tabular data manipulation:

(A) NumPy

(B) pandas ✓

(C) matplotlib

(D) requests

Answer: (B) pandas

Authenticity: Community Reported

Q10. Python structure with unique unordered elements:

(A) list

(B) tuple

(C) set ✓

(D) dict

Answer: (C) set

Authenticity: Community Reported

Section 4 - SQL

Problem: Second Highest Salary

Find the second highest distinct salary. (Explicitly reported in BI3 technical round.)

```
SELECT MAX(salary) AS second_highest
FROM employee
WHERE salary < (SELECT MAX(salary) FROM employee);
```

Expected Output

```
second_highest
42000
```

Section 5 - Coding / Python

Problem 1: Two Sum

Difficulty: *Easy* | Pattern: *Hash Map* | Authenticity: *Frequently Repeated*

Problem Statement

Return indices of the two numbers adding up to the target.

Input Format

Line 1: array. Line 2: target.

Output Format

Two indices, space-separated.

Constraints

$2 \leq n \leq 10^4$

Sample Input

```
3 2 4
6
```

Sample Output

```
1 2
```

Python Solution

```
def two_sum(nums, target):
    seen = {}
    for i, x in enumerate(nums):
        if target - x in seen:
            return (seen[target - x], i)
        seen[x] = i

i, j = two_sum([3, 2, 4], 6)
print(i, j)
```

Step-by-Step Explanation

Store indices; when the complement appears, return the pair. $2 + 4 = 6$.

Dry Run

at 4, complement 2 seen -> (1, 2)

Time Complexity: **$O(N)$** | Space Complexity: **$O(N)$**

Problem 2: Happy Number

Difficulty: **Easy** | Pattern: **Cycle Detection (Set)** | Authenticity: **Verified Previous-Year**

Problem Statement

A happy number reaches 1 by repeatedly summing the squares of its digits; otherwise it loops. Decide if n is happy. (Explicitly reported in BI3 technical round.)

Input Format

A single integer n .

Output Format

True or False.

Constraints

$1 \leq n \leq 2^{31} - 1$

Sample Input

19

Sample Output

True

Python Solution

```
def is_happy(n):
    seen = set()
    while n != 1 and n not in seen:
        seen.add(n)
        n = sum(int(d) ** 2 for d in str(n))
    return n == 1

print(is_happy(19))
```

Step-by-Step Explanation

Repeatedly sum digit squares; a set detects loops. $19 \rightarrow 82 \rightarrow 68 \rightarrow 100 \rightarrow 1$.

Dry Run

$19 \rightarrow 82 \rightarrow 68 \rightarrow 100 \rightarrow 1 \rightarrow \text{True}$

Time Complexity: **$O(\log n)/\text{step}$** | Space Complexity: **$O(\log n)$**

BI3 Technologies

Full-Length Mock Test 1

Applied Role(s):

Data Engineer - Trainee

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.